

AI-driven prediction of drug activity against *Toxoplasma gondii*: Data augmentation and deep neural networks for limited datasets

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ABSTRACT

Toxoplasmosis, caused by *Toxoplasma gondii* (*T. gondii*), is a serious global health concern, particularly in immunocompromised individuals. Inhibiting the enzyme TgDHFR is a promising strategy for developing treatments. This Artificial Intelligence (AI)-driven Quantitative Structure-Activity Relationship (QSAR) study applies deep neural networks (DNNs) to predict pIC₅₀ values for potential inhibitors, using 2D and 3D molecular descriptors and fingerprints. To address training data limitations, we introduced a novel methodology combining targeted descriptor selection, Gaussian noise-based data augmentation, and an ensemble of DNNs. This approach significantly enhanced model performance, increasing the R² from 0.75 with the original dataset to 0.85. The model was further validated using two FDA-approved drugs for *T. gondii* treatment—pyrimethamine and trimethoprim—yielding relative errors of 3.35 % and 2.15 % in pIC₅₀ predictions compared to experimental values. Finally, the model was applied to screen FDA-approved drugs after filtering out molecules that did not align with the characteristics of the training dataset. The predicted pIC₅₀ values were further used to calculate ligand efficiency (LE), binding efficiency index (BEI), lipophilic ligand efficiency (LLE), and surface efficiency index (SEI), identifying the most promising TgDHFR inhibitors for further investigation. By leveraging AI and data augmentation approach, this study provides a powerful tool for pIC₅₀ predictions of TgDHFR inhibitors, which can be adapted to other systems.

1. Introduction

T. gondii is a widespread protozoan parasite that infects various warm-blooded animals, including humans. It is a food and waterborne pathogen and one of the most prevalent parasitic infections globally. Infected humans can show influenza-like symptoms like muscle aches, swollen lymph nodes and eye problems within weeks to months. Neurological symptoms can be developed among patients with weakened immune systems. The acute form may cause fever, pneumonia, and even death [1,2].

There is a challenge to develop an effective, inexpensive, and practical treatment against toxoplasmosis with reduced toxicity, enhanced penetration of the brain and eyes, reduced duration, and safety for pregnancy. Human toxoplasmosis has been treated with drugs that are effective against other parasites. Examples include sulfadiazine, pyrimethamine, autocon, trimethoprim, and sulfamethoxazole [1–3].

However, due to the parasite's complex life cycle, the development of resistance and treatment failure in Toxoplasma chorioretinitis, encephalitis, and congenital infections is increasing and needs further investigation. The current treatments can be long-term as the current clinical drugs cannot eliminate the cystic stage of *T. gondii*. The approved drugs cannot clear chronic infections in the human body; hence, the major population of humans is at risk of severe consequences. Also, there is a need for drugs with fewer and milder side effects instead of old drugs that do not respond to the treatment and have high side effects [2,4].

The proteins *T. gondii* Dihydrofolate Reductase (TgDHFR), Prolyl-tRNA Synthetase (TgPRS), and Calcium-Dependent Protein Kinase 1 (TgCDPK1) are among the proteins that are crucial for developing treatments against toxoplasmosis [5]. Numerous studies have explored the development of specific inhibitors targeting TgCDPK1, demonstrating efficacy in vitro and animal models, underscoring their potential as anti-toxoplasma drugs [5]. Our study mainly focuses on TgDHFR,

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which plays a vital role in the folate metabolic pathway. The 2,4-diaminopyrimidine derivative pyrimethamine, a widely used DHFR, is an FDA-approved inhibitor for the treatment of toxoplasmosis. A safer treatment for toxoplasmosis can be obtained by increasing the selectivity and potency of DHFR inhibitors [6]. This has been used in toxoplasmosis treatment for decades, yet the therapeutic improvements are long overdue [7]. Hence, advanced drug discovery methodologies are required to identify compounds with enhanced potency and selectivity for TgDHFR.

To date, a range of methods in computer-aided drug discovery have contributed mainly to advancing drugs [8,9]. Drug response prediction is imperative in developing valid drug discovery methods. The IC_{50} is a metric used to measure the cell viability of half-maximal inhibitory concentration and predict the drug response. Various drug response prediction models have emerged due to experiments to develop drugs targeting different diseases [8,10–16]. The drug response has been predominantly predicted using Machine learning (ML) models despite the advances and advantages of deep learning (DL). Recent advancements in machine learning (ML) and deep learning (DL) models have significantly transformed the field of drug discovery, particularly in predicting inhibitory concentration (IC_{50}) values [17–20]. DL architectures, including graph neural networks (GNNs), recurrent neural networks (RNNs), and transformer-based models, have demonstrated substantial success in various applications, such as virtual screening [21, 22], binding affinity prediction [23], and activity prediction [24]. Prominent DL frameworks like MoleculeNet [25], ChemProp [23], or the DeepIC50 [26] have achieved high predictive accuracy in modeling biological activity. However, these models are predominantly trained and optimized on large, well-established datasets focusing on human targets or widely studied proteins, such as cancer-related ones. So, their performance can be limited when applied to datasets beyond their original training scope. Notably, MoleculeNet and ChemProp struggle to achieve reliable predictions on small datasets, highlighting their limited generalizability in data-scarce scenarios [27,28]. Unfortunately, in our current work, we are dealing with a limited heterogeneous dataset of TgDHFR, comprising fewer than 1000 entries, which necessitates non-trivial strategies to address the limitations of existing DL models on small datasets.

There are several models in the literature for predicting DHFR inhibition, but they are predominantly trained on small, homologous datasets of inhibitors with specific heterocyclic cores, such as quinazolines or pyrimidines (Table 1). Only a few studies have been performed with inhibition data from the *T. gondii* of DHFR, and these existing models often suffer from poor generalization. For example, Mattioni and Jurs [29] developed artificial neural network models to predict pIC_{50} values for the three types of DHFR: *Rat liver*, *Pneumocystis carinii*, and *Toxoplasma gondii* Dihydrofolate Reductase. In this work, they used 345

compounds. The model performs well on the training set ($R^2=0.917$) and well during cross-validation ($R^2=0.852$). However, there is a significant performance drop on the test set ($R^2 = 0.626$), suggesting that the model has limited generalization to unseen data.

Therefore, it is essential to develop a methodology that addresses the challenges posed by limited heterogeneous data and establishes a model capable of delivering robust performance in molecular property prediction across diverse chemical compounds. In this project, we identified potential TgDHFR inhibitors via FDA-approved drug screening, employing a deep neural networks (DNNs) model for pIC_{50} prediction. The study is structured into four key stages: (i) a baseline DNN model was initially developed using a limited training dataset, which was insufficient for broad generalization. (ii) The second stage focused on enhancing the model by analyzing the impact of different molecular descriptors, identifying the most significant features, augmenting the training data, and applying an ensemble of DNNs to improve performance. (iii) We prepared the FDA-approved drug dataset in the third stage. Analysis of some main molecular characteristics revealed significant differences between the FDA dataset and our training set, which could reduce model reliability. To address this, we trained a machine learning model to identify outliers in the FDA dataset and then filtered out compounds that did not align with the training data characteristics. (iv) pIC_{50} values were predicted for the filtered FDA compounds in the final stage. Key ligand efficiency parameters, such as LE, BEI, LLE, and SEI, were also calculated to identify the most promising inhibitors for further evaluation.

2. Training dataset

2.1. Database

In this work, we utilized two databases to construct a training set for the DNN model. The first database (BindingDB_All_202311_tsv.zip) was sourced from the UC San Diego SKAGGS School of Pharmacy and Pharmaceutical Sciences (BindingDB) [35]. This database is a vast collection of 2874,595 entries, each representing a record of drug-protein interaction. It encompasses a wide array of columns (194 in total) that detail various aspects of these interactions. Key columns include ligand SMILES and protein amino acid sequences, which capture essential chemical and biological information. The dataset also provides experimentally measured data such as IC_{50} , K_i , K_d , and EC_{50} values. Notably, it includes detailed identifiers and sequences for both ligands and target proteins, spanning sources like PubChem, ChEMBL, and UniProt. The second database, ChEMBL [36], provided additional data, including ligand SMILES and related IC_{50} values for TgDHFR protein. However, after merging these two datasets and removing all duplicates, the final database has 873 samples of unique compounds.

2.2. Target (pIC_{50})

A detailed examination of these two databases revealed multiple IC_{50} values for the same ligand, indicating potential variations in experimental conditions or measurements. To address the issue of multiple IC_{50} values for the same ligand in the databases, a systematic approach was implemented using the Interquartile Range (IQR) method to remove outliers [37]:

$$IQR = Q3 - Q1 \quad (1)$$

$$LB = Q1 - 1.5 \times IQR \quad (2)$$

$$UB = Q3 + 1.5 \times IQR \quad (3)$$

Where IQR - interquartile range, $Q1$ - first quartile (25th percentile), $Q3$ - first quartile (75th percentile), LB - lower bound, UB - upper bound. Initially, the data was grouped by the ligand's CID number, and the IQR

Table 1
Computational Approaches for Predicting DHFR Inhibition: Models, Inhibitors, and Protein Targets.

Model	Inhibitors	Protein Target	Reference
ANN	Nonclassical Antifolates	Pneumocystis carinii DHFR, Toxoplasma gondii DHFR	Ref.[30]
SVM	Pyrimidines	DHFR*	Ref.[31]
Bilinear Model	Pyrimidines	Bacterial and Avian DHFR	Ref.[32]
ANN	Triazines	DHFR*	Ref.[33]
CoMFA	Pyrimidines	Rat liver DHFR, Toxoplasma gondii DHFR	Ref.[34]
ANN	Pyrimidines, Pteridines, Quinazolines etc.	Pneumocystis carinii DHFR, Toxoplasma gondii DHFR, Rat liver DHFR	Ref.[29]

*DHFR refers to a general dihydrofolate reductase enzyme without species-specific designation.

method was applied to each group to filter out outliers in the IC_{50} values. After outliers were removed, the average IC_{50} value was calculated for each ligand group. This averaged value was then used to replace the original IC_{50} values, ensuring a consistent and representative IC_{50} value for each ligand across the dataset. Finally, a logarithmic transformation was applied to the IC_{50} (nM) values to convert them into pIC_{50} values, as defined by the following equation [38]:

$$pIC_{50} = -\log_{10}(IC_{50} (M)) \quad (4)$$

The pIC_{50} transformation is essential because it standardizes the IC_{50} values on a logarithmic scale, making them more manageable and reducing the impact of extreme values. This transformation also improves the interpretability and comparability of the data, as pIC_{50} values are typically within a narrower range and are more suitable for regression modeling. The pIC_{50} value, which inversely correlates with the ligand's potency, serves as the target variable in this study.

2.3. Ligand descriptors

The SMILES were utilized to evaluate the descriptors and fingerprints. Descriptor calculations were performed using PaDELpy, a Python wrapper for the PaDEL-Descriptor software [39]. This software allowed us to calculate 1875 descriptors (1444 1D, 2D descriptors, and 431 3D descriptors) and the PubChem fingerprint (FP) of 881 bits long (total of 2756 descriptors for each ligand). Notice: since 1D and 2D descriptors are computed from the molecular graph (a 2D structure), they are grouped under one label (2D), simplifying the categorization and avoiding confusion.

To calculate 3D descriptors, first, we employed the RDKit library [40] to generate and optimize molecular conformers for a set of unique SMILES strings. Each SMILES string was converted into a molecular object, followed by the addition of hydrogen atoms to complete the valence. For each molecule, ten conformers were generated using the ETKDGv3 method [41], which accounts for the geometrical constraints in conformer generation. This method combines distance geometry and experimental torsion angles, generating realistic conformers while maintaining stereochemistry and avoiding unrealistic bond angles, which is ideal for 3D descriptor calculation or docking. The conformers were then optimized using the MMFF94s force field [42] to achieve energy minimization. This force field was optimized for small organic molecules, offering more accurate electrostatic interaction treatment and improved energy minimization. The conformer with the lowest energy was selected, annotated with its corresponding SMILES string, and written to an SDF (Structure-Data File). This process ensured that each molecule was represented in its most stable conformational state in the resulting SDF file, enabling accurate calculation of 3D descriptors for further analysis.

3. Methodic

3.1. Feature engineering

This study employed a series of preprocessing steps to prepare the dataset for machine learning analysis, leveraging tools from the *feature-engine*[43] and *scikit-learn*[44] libraries. In the first step, the data was partitioned into training and testing subsets using the *train_test_split* function from *scikit-learn*. Because our training dataset is not large, we allocated 15 % of the data for the testing set (*test_size*=0.15), which should provide a good balance between training and evaluation. A random seed (*random_state*=42) was set to guarantee the reproducibility of the results. In the preprocessing phase, we addressed missing values using the *MeanMedianImputer* from the *feature-engine* library, applying median imputation to ensure robust handling of gaps in the data. To manage outliers, we employed the *Winsorizer* transformer, capping extreme values at both tails of the *Gaussian* distribution, thereby stabilizing variance and improving data robustness. We further applied

the *Yeo-Johnson* transformation to normalize feature distributions, enhancing variance stability and ensuring the data was well-suited for modeling. For feature scaling, the *StandardScaler* from *scikit-learn* was utilized to standardize all features, preventing any single feature from disproportionately influencing the model due to scale differences. These comprehensive preprocessing steps improved data quality and consistency, providing optimal conditions for effective machine learning model development and evaluation.

3.2. Deep-learning model parameters

A deep-learning regression model was developed using TensorFlow [45,46] and Keras [47] in Python to predict the target variable, pIC_{50} . The model architecture is a fully connected feedforward neural network [48], a multi-layer perceptron consisting of four dense layers with 128, 64, 32, and 16 neurons, each utilizing the ReLU activation function. Batch Normalization layers were incorporated after each dense layer to improve generalization and reduce the risk of overfitting. These layers standardize the inputs to each subsequent layer, accelerating training and stabilizing learning by reducing internal covariate shifts.

To further combat overfitting, Dropout layers were used with progressively reduced dropout rates, starting at 0.4 in the first hidden layer and tapering to 0.2 in the final layers. This technique of stochastic regularization randomly omits a fraction of neurons during training, preventing the model from becoming overly dependent on any single neuron and enhancing its robustness. The model was compiled with the Adam optimizer, set at an initial learning rate of 0.0007, providing a balanced approach to convergence speed and accuracy. Mean Squared Error (MSE) was selected as the loss function suitable for regression tasks. Mean Absolute Error (MAE) was used as an additional performance metric to assess predictive accuracy further.

To optimize the training process, the *ReduceLROnPlateau* callback was employed, which dynamically reduces the learning rate by a factor of 0.2 if no improvement in validation loss is observed for 20 epochs, with a minimum learning rate set at 0.000001. The *EarlyStopping* callback was applied to monitor validation loss, terminate training after 30 consecutive epochs without improvement (or 60 epochs for the augmented dataset), and automatically restore the best model weights achieved during training. The model was trained on the dataset with a batch size of 32 over 200 epochs, using *X_test* and *y_test* as the validation set to monitor performance.

3.3. DNN model ensemble

An ensemble approach was implemented to enhance robustness and reduce prediction variability by averaging the predictions from multiple runs of the neural network model. Specifically, the model was trained and evaluated five times, with predictions from each run being stored and averaged to produce the final output. This ensemble method mitigates the influence of stochastic elements, such as random weight initialization and batch selection, resulting in more stable and reliable predictions. The MAE and R^2 scores were computed for each run, and the overall performance metrics were determined from the averaged predictions. This comprehensive approach provided a robust evaluation of the model's effectiveness, as illustrated in Fig. 1.

3.4. Feature importance calculation

We applied the Permutation Feature Importance method [49] to quantify the contribution of individual features in our deep learning model for predicting pIC_{50} values. This approach involves shuffling each feature in the test dataset and measuring the change in model performance, using the negative mean squared error as the metric. This study also investigated the impact of features with varying levels of importance on model accuracy.

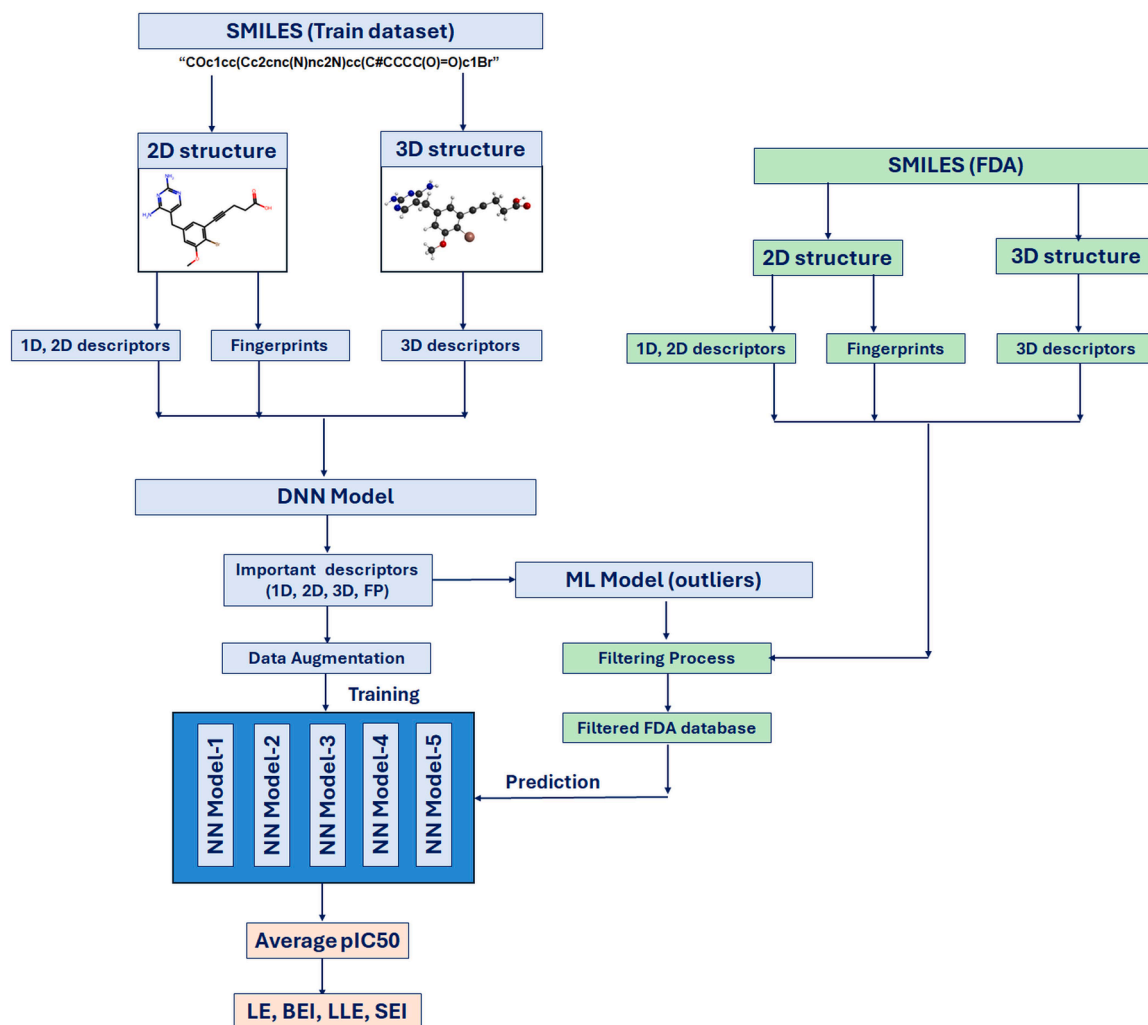


Fig. 1. Workflow for pIC_{50} prediction using ensemble neural network models. An example ligand is 4-pentanoic acid. Blue color – is the training dataset, green color – is the FDA dataset, and orange – is the results.

3.5. Data augmentation

Due to limited IC_{50} data for the protein of interest, synthetic data were generated using Gaussian noise [50], a widely used data augmentation technique. Gaussian noise follows a normal distribution, introducing variability into the feature training set and not to target values. Two noise levels were applied: first, with a standard deviation of 0.01, followed by a noise with a smaller standard deviation of 0.001. This process created two augmented versions of each data point, enhancing the training dataset and improving the model's robustness against variability. To ensure a reliable model performance evaluation, we employed a distinct data-splitting strategy for the augmented model that prioritized testing on real, non-augmented data. Unlike the approach used for the non-augmented models, the augmented model required a different approach to reduce potential biases introduced by synthetic data.

Before augmentation, 15 % of the original dataset was set aside as an independent test set, ensuring that this portion remained completely unseen during augmentation and model training. The remaining original dataset was used to generate augmented samples, thereby increasing the size and diversity of the training set. After augmentation, the model was trained on the augmented dataset, and its performance was evaluated using the 15 % independent test set of original, non-augmented data. This strategy ensures that the model's generalization ability is assessed on unseen, real-world data, avoiding the risk of overestimation of

accuracy that can occur when testing on augmented data. By keeping the test set free from any augmented samples, we aimed to obtain a more realistic and unbiased measure of the model's practical applicability.

3.6. Ligand efficiency metrics

In addition to the pIC_{50} , metrics such as Ligand Efficiency, Binding Efficiency Index, Lipophilic Ligand Efficiency, and Surface Efficiency Index provide crucial insights into the drug-like properties of a compound. These metrics offer a more nuanced evaluation of a ligand's potency relative to its molecular size, lipophilicity, and polar surface area. Ligand Efficiency (LE) helps normalize the binding affinity to the size of the molecule, making it helpful in comparing compounds of different molecular weights. It measures how efficiently a molecule utilizes its atoms to bind to a target protein [51–53]:

$$LE = 1.37 \times \frac{pIC_{50}}{HAC} \quad (5)$$

where HAC is the number of non-hydrogen atoms in the molecule (heavy atom count).

The Binding Efficiency Index (BEI) evaluates efficiency relative to molecular weight [51]:

$$BEI = \frac{pIC_{50}}{MW} \times 1000 \quad (6)$$

where MW is the molecular weight of the molecule in kilodaltons (kDa). A higher BEI value indicates a more efficient ligand. This means the molecule has a stronger affinity for the target protein relative to its size.

Another valuable tool for drug discovery is Lipophilic Ligand Efficiency (LLE), which provides a quantitative measure of a molecule's efficiency by considering its potency and lipophilicity [53]:

$$LLE = pIC_{50} - \log P \quad (7)$$

where $\log P$ is the logarithm of the partition coefficient between octanol and water (the related calculated descriptor is XlogP), a measure of lipophilicity that influences a molecule's absorption, distribution, metabolism, and toxicity. A higher LLE value indicates a more efficient ligand: a high pIC_{50} value suggests strong binding, while a low $\log P$ value suggests lower lipophilicity, which can benefit drug-like properties.

Surface Efficiency Index (SEI) is a metric used in drug discovery to evaluate the efficiency of a molecule's binding to a target protein based on its surface area. It measures how efficiently a molecule utilizes its surface area to interact with the protein [51,53,54]:

$$SEI = \frac{pIC_{50}}{PSA} \times 100 \quad (8)$$

where PSA is polar surface area ($\text{\AA}^2$), a higher SEI value indicates a more efficient ligand. This means the molecule has a stronger affinity for the target protein relative to its surface area. In other words, it is making the most of its surface area regarding binding potency.

Together, these metrics complement pIC_{50} by providing a more comprehensive assessment of a compound's efficiency and drug-likeness, guiding the optimization of leads in drug discovery.

4. Results and discussion

4.1. Target transformation effect

The variable IC_{50} , a key measure in pharmacological studies representing the concentration of a drug that is required for 50 % inhibition in vitro. In the database, raw IC_{50} values are in nanomolar (nM). The histogram of IC_{50} values shows a highly skewed distribution, with most values clustered at the lower end (Fig. 2a). This severe skewness is

confirmed by the Q-Q plot (Fig. 2a'), where the data points deviate significantly from the red normality line, especially in the tails. The Shapiro-Wilk test statistic and an extremely low p-value (Fig. 2a') further indicate that the IC_{50} values are far from normally distributed, making them unsuitable for many statistical analyses that assume normality.

To address this skewness and make the data more useful for analysis, a logarithmic transformation was applied to the IC_{50} (nM) values using Eq. (4). The histogram of the pIC_{50} values (Fig. 2b) shows a more symmetric distribution, although it still exhibits some skewness. The Q-Q plot (Fig. 2b') indicates an improvement in normality, with the data points aligning more closely to the red line than the raw IC_{50} values. However, deviations are still noticeable, particularly at the tails, as reflected in the Shapiro-Wilk test statistic and a p-value, which suggests that while the data is closer to normal, it still does not meet the normality assumption.

Further improvement was obtained by applying a Quantile Transformation (QT) to the pIC_{50} values of the entire dataset to force the distribution into a more normal shape (QT- pIC_{50}). The histogram (Fig. 2c) of the transformed data shows a distribution closely resembling a standard normal distribution with a precise bell shape. The Q-Q plot (Fig. 2c') demonstrates significant alignment of the data points with the red normality line, indicating a much closer fit to a normal distribution. The Shapiro-Wilk test results suggest that the data is still not perfectly normal but much closer than previous transformations.

Outliers were filtered from the QT- pIC_{50} data using the Z-score test to refine the normality further. The filtered data's histogram (Fig. 2d) shows an even more refined normal distribution, with outliers removed, resulting in a tighter bell curve. The Q-Q plot (Fig. 2d') shows almost perfect alignment with the red normality line, and the Shapiro-Wilk test statistic is 0.9997 with a p-value of 0.99, indicating that the data is now normally distributed. This final step, achieved through Z-score-based outlier removal, ensures that the data is well-suited for analyses that require normality, eliminating the deviations and outliers that could have influenced the results.

The model performance was tested and compared for these two targets, pIC_{50} and QT- pIC_{50} , both with filtered outliers. This approach will help determine whether the additional transformation and outlier filtering steps significantly improve the model's predictive accuracy.

The model performance was evaluated for both targets, utilizing all

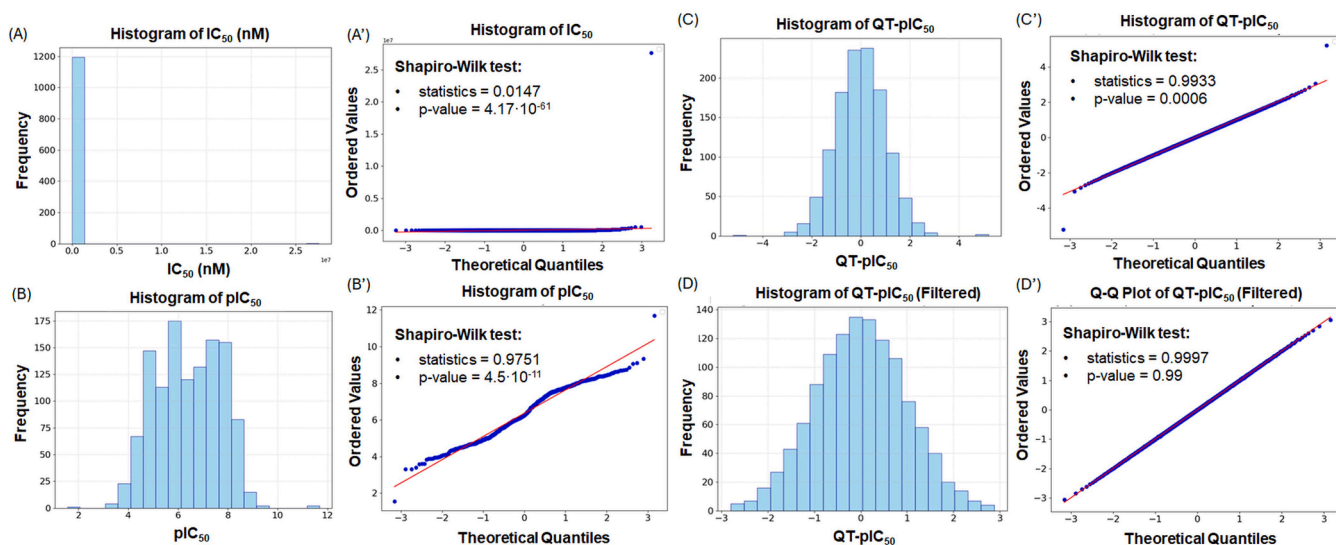


Fig. 2. Histograms (A-D) and Q-Q plots (A'-D') illustrate the distribution and normality of IC_{50} and pIC_{50} values at different transformation stages. Panels (A-A') present the raw IC_{50} values. (B-B') show the pIC_{50} values calculated from IC_{50} using Eq. (4). (C-C') display the results after applying a Quantile Transformation to the pIC_{50} values. (D-D') present the results after applying the Quantile Transformation and filtering outliers. The Q-Q plots include the Shapiro-Wilk test statistics and p-values for each dataset.

ligand descriptors. The comparison between the pIC_{50} and QT-pIC_{50} models reveals that the model trained on the pIC_{50} target demonstrates superior overall performance (Fig. 3). Specifically, the pIC_{50} model achieved a higher R^2 score of 0.75, indicating a better ability to explain the variance in the data compared to the QT-pIC_{50} model, which had an R^2 of 0.69. Additionally, the Training vs. Validation MAE curves for the QT-pIC_{50} model displayed more pronounced oscillations, likely due to the quantile transformation applied to the data. While this transformation can improve data normality, it may also introduce inconsistencies in how the model learns, leading to increased variability in training and validation performance (Fig. 3c and c'). Also, the additional transformation might result in a less stable and interpretable representation of the underlying data patterns, making it more challenging for the model to converge on a stable solution, hence the observed oscillations.

Therefore, the comparison between QT-pIC_{50} and pIC_{50} results highlights that while quantile normalization can improve data distribution, it may also introduce challenges in achieving stable and consistent model performance. As a result, the QT-pIC_{50} target was excluded from further analysis, and all subsequent studies were conducted using the pIC_{50} target, demonstrating superior performance and excellent stability.

4.2. Ligand descriptors effect

To assess the impact of different ligand descriptors on the performance of the deep learning regression model for predicting pIC_{50} , we evaluated four types of input data: 2D descriptors only (2D model), 2D descriptors with fingerprints (2D/FP model), 2D and 3D descriptors together (2D/3D model), and all descriptors 2D, 3D and fingerprint (2D/

3D/FP model). The goal was to evaluate the impact of different combinations of descriptors on the model's performance. The results were summarized in Fig. 4, which displays the box plot of relative error (%) for each model configuration, along with their respective R^2 values.

The model with 2D descriptors only achieved a median relative error of 0.72 % (Fig. 4). The narrow error distribution indicates reasonable accuracy and consistency, though the model shows some variability in its predictions. Adding fingerprint descriptors to the 2D model (resulting in the 2D/FP model) increased the median relative error to 0.96, but also improved the R^2 value, indicating better overall model performance.

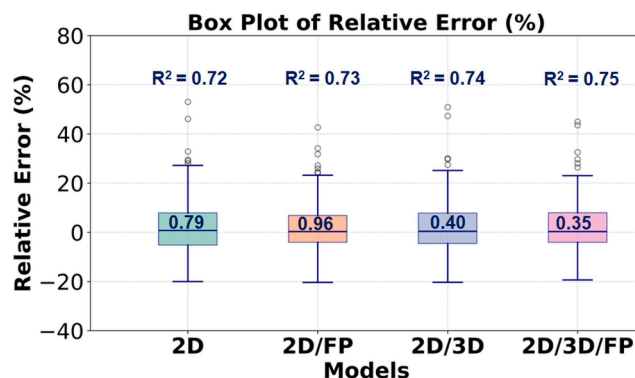


Fig. 4. Box plot of relative error (%) for comparison of DLRM performance with different ligand descriptors: 2D, 2D/FP, 2D/3D, and 2D/3D/FP. The R^2 values for each model configuration are shown above the corresponding box plots.

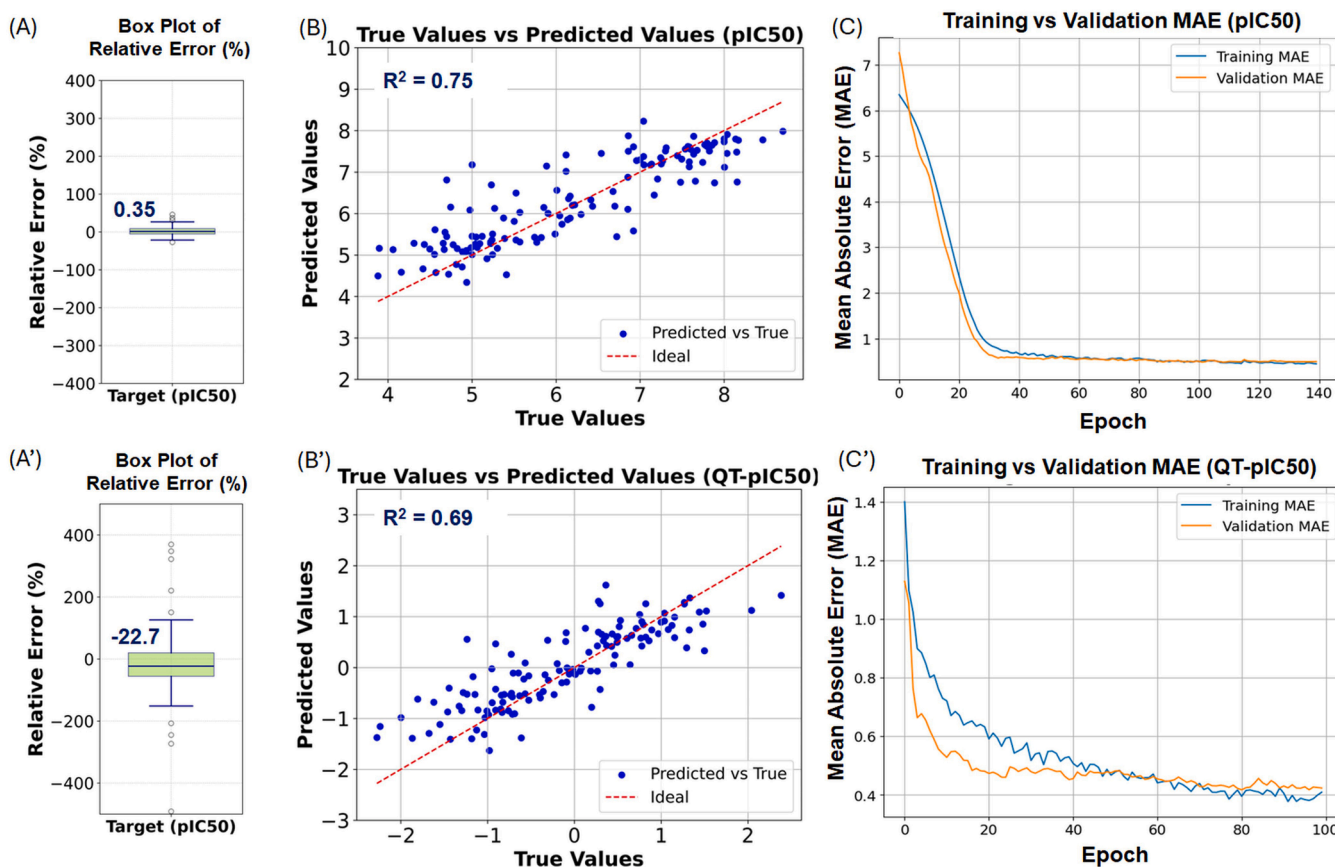


Fig. 3. Comparison of the performance for pIC_{50} and QT-pIC_{50} models. Model pIC_{50} : (A) boxplot of relative error, (B) scatter plot of actual values vs predicted values, (C) training vs validation MAE curve. Model QT-pIC_{50} : (A') boxplot of relative error, (B') scatter plot of actual values vs predicted values, (C') training vs validation MAE curve.

Incorporating 3D descriptors into the 2D/3D model further reduced the median relative error to 0.40 %, highlighting the importance of 3D structural information in refining predictions. Finally, the 2D/3D/FP model with all three descriptor types achieved the lowest median relative error of 0.35 % and the highest R^2 value of 0.75, demonstrating an optimal balance between accuracy and variability.

Statistical analysis revealed that the differences in median relative errors between the 2D model and other models were not statistically significant. Despite this, the 2D/3D/FP model consistently demonstrated the best overall performance, achieving the highest R^2 and the smallest median error. To further enhance its performance, feature importance analysis could identify and eliminate less relevant or detrimental features, thereby reducing the median error and improving the model's predictive accuracy.

4.3. Features importance

The importance of features in a 2D/3D/FP model was assessed using the Permutation Importance (PI) approach. The inspection module evaluates each feature's importance by measuring the change in model performance, quantified by the negative mean squared error when the feature values are randomly shuffled. This process is repeated five times ($n_{\text{repeats}}=5$) to ensure robust estimates. The resulting importance scores represent the average impact of permuting each feature on the model's performance.

The results indicate that certain features positively impact the model's predictions, while others have zero or even negative importance scores. Features with positive importance are crucial for the model's accuracy, whereas those with zero importance do not affect the model's performance and may be redundant. Interestingly, some features exhibit negative importance scores, suggesting that their presence could introduce noise or redundancy, potentially degrading model performance. These results highlight the necessity of refining the feature set by removing or reconsidering features with zero or negative importance to enhance the model's overall accuracy and efficiency. Fig. 5a illustrates the distribution of ligand features positively impacting model performance. The pie chart shows that 2D descriptors comprise the most significant portion of these essential features, accounting for 58.1 %. FP (molecular fingerprint) and 3D descriptors contribute similarly to the features, though their amount is smaller than that of 2D descriptors.

Fig. 6 illustrates the distribution of critical descriptors across three categories—2D, 3D, and Fingerprint (FP)—based on their impact in a pIC_{50} prediction model. Each pie chart reflects the percentage contribution of different descriptor types within each category. In the 2D descriptors category, 'AutocorrelationDescriptor' is predominant, accounting for 33.3 % of the total, followed by 'ElectropologicalStateAtomTypeDescriptor' at 20.0 %. The prominence of these descriptors indicates that structural and electronic features are crucial in determining the binding affinity of molecules to the target protein. In the 3D descriptors category, 'RDFDescriptors' (radial distribution function descriptors) are the most significant (59.3 %), emphasizing the importance of atomic distance distribution in understanding molecular structure and properties. For Fingerprint (FP) descriptors, 'SMARTS Patterns' (46.0 %) and 'SMARTS Patterns 2' (26.5 %) are the most dominant. These two descriptor types differ in complexity and specificity: 'SMARTS Patterns 2' target complex substructures with precise bond and aromaticity matching, whereas 'SMARTS Patterns' are more flexible, detecting broader substructures with less specificity. Both are essential for capturing different levels of molecular detail; 'SMARTS Patterns 2' provides critical insights into specific chemical features, while 'SMARTS Patterns' provides broader, general analyses. Together, they deliver a comprehensive approach to chemical structure analysis. The accompanying CSV file, available in the [Supporting Information \(SI\)](#), provides detailed insights into the impact of each descriptor, including its class, type, and description. This information is valuable for refining the model and identifying the molecular features most

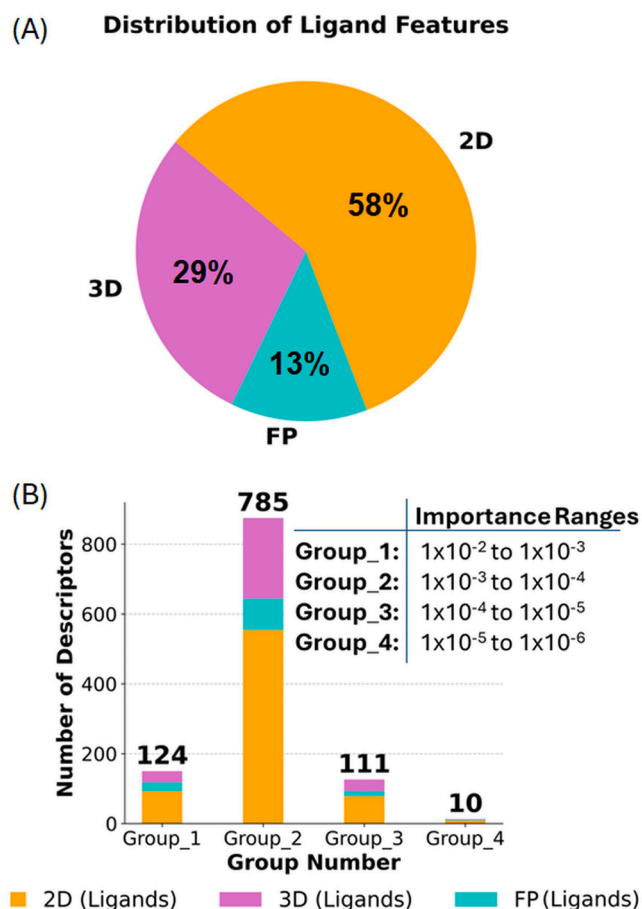


Fig. 5. (A) Distribution of descriptor class among features with a positive impact only (total of 1030 features). (B) Percentage composition of descriptors in each important score group for ligand features (model 2D/3D/FP).

significantly contribute to ligand activity.

To better understand the influence of different descriptors on model performance, the features with a positive impact were divided into four groups based on their importance values. Group 1 includes values ranging from 10^{-2} to 10^{-3} , Group 2 covers the range from 10^{-3} to 10^{-4} , Group 3 contains values between 10^{-4} to 10^{-5} , and Group 4 represents the lowest range, spanning from 10^{-5} to 10^{-6} (Fig. 5b). Group 1, which includes the most impactful features, predominantly comprises 2D descriptors, underscoring their critical role in enhancing model accuracy. In Group 2, the largest group by far, 2D descriptors continue to dominate. However, this group also includes a noticeable presence of FP and 3D descriptors, indicating that these contribute significantly to model performance, though to a lesser extent. Group 3 shows a further reduction in the number of 2D descriptors but maintains a substantial count of FP and 3D descriptors, suggesting that these descriptors become increasingly relevant as the importance score decreases.

In this study, we further investigated the impact of different groups of molecular descriptors on the performance of a deep-learning regression model. We observed how excluding specific groups influenced model performance by systematically removing descriptor groups from the model with all descriptors. Notably, starting with the model using all descriptors, the median relative error was 0.35 % with an R^2 value of 0.75, providing a strong baseline of performance (Fig. 7a-e). When we tested the model without certain descriptor groups, the results showed varying impacts on performance. The Group 1, 2, 3, 4 model (all features with positive impact only) exhibited the highest R^2 value of 0.82, with a slightly improved median relative error of 0.32 %. This indicates that the model maintained high predictive power with fewer descriptors but

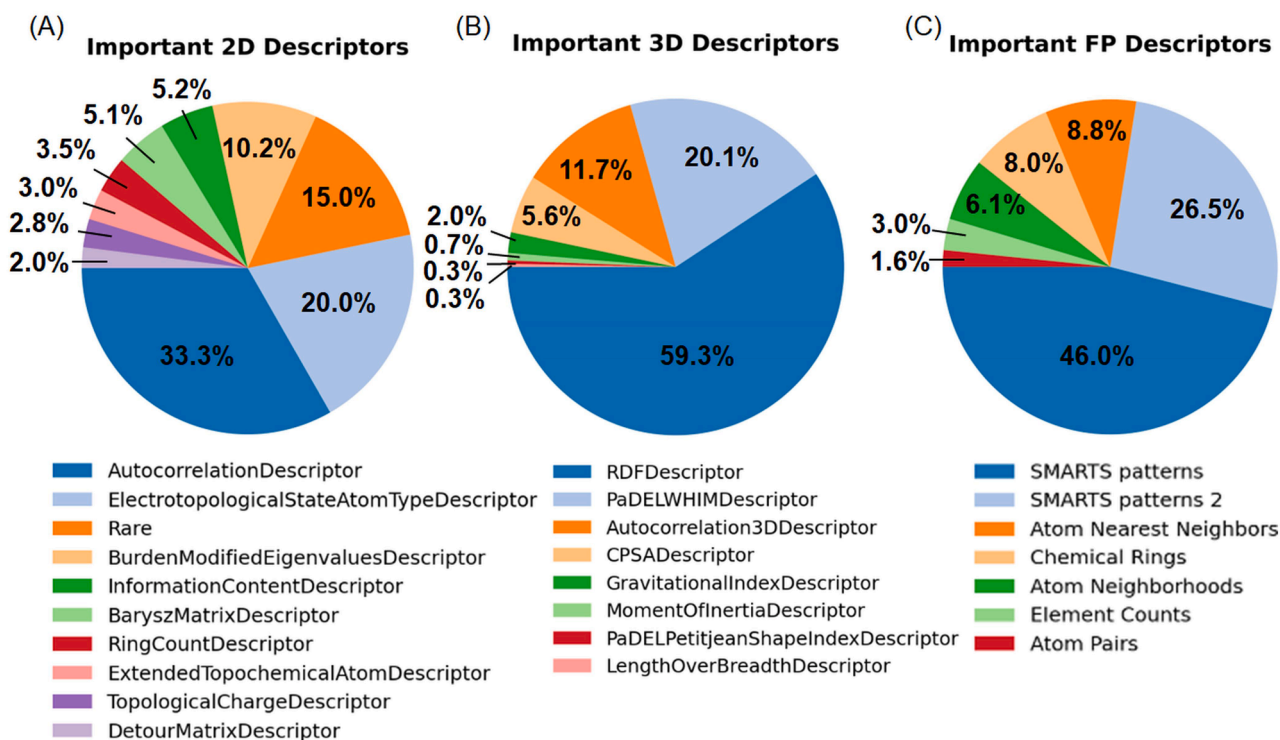


Fig. 6. Composition of 2D, 3D, and FP descriptors with positive impact only: (A) contribution of the important descriptors in class 2D ('Rare' group contents all descriptors with a population less than 2.0 %, see Table S1), (B) contribution of the important descriptors in class 3D, and (C) contribution of the important descriptors in class FP.

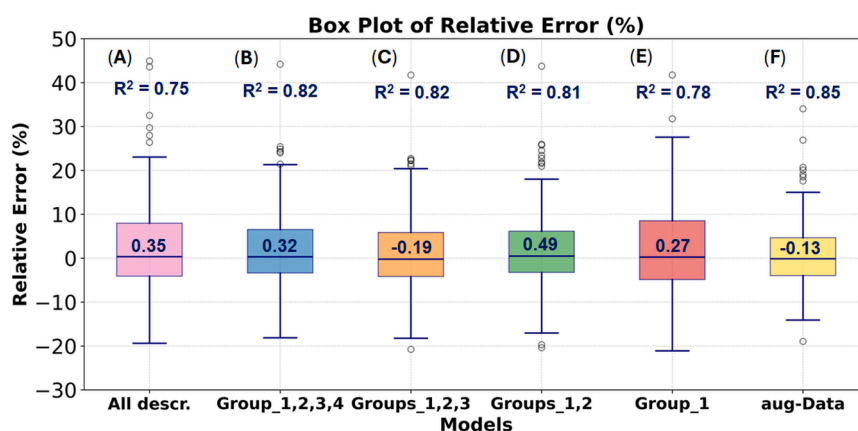


Fig. 7. Box plot of relative error (%) for comparison of DLRM performance with descriptors from. (A) All descriptors were used; (B)–(E) descriptors with positive impact were considered; and (F) model was obtained for augmented data with descriptors of groups 1–3.

was still accurate and consistent. Removing Group 4 descriptors (forming the Group 1, 2, 3 models) further improved the median relative error to -0.19% , representing the lowest error across all models, while the R^2 value remained high at 0.82. This suggests that some descriptors might not only be unnecessary but could even introduce slight bias when included, as seen in the negative error shift after removal.

However, as we removed additional descriptor groups, the performance slightly declined. The Group 1, 2 model demonstrated a median relative error of 0.49% , and while its R^2 value remained strong at 0.81, the error distribution became wider, indicating more prediction variability. Finally, the Group 1 model showed a further decline, with a median relative error of 0.27% and the lowest R^2 value of 0.78, emphasizing the importance of retaining multiple descriptor groups for maintaining accuracy.

These findings show that the best results were achieved with the

Group 1, 2, 3 model, balancing high explained variance with improved prediction accuracy and reduced error variability. Including these groups captures enough molecular complexity for accurate predictions, highlighting the need for a balanced selection of descriptors to maintain both accuracy and consistency.

Further model improvements were achieved through data augmentation, expanding the training dataset to 2355 samples. This larger dataset, combined with features from the three most important descriptor groups, resulted in significant performance gains. The R^2 on the validation set, which consisted of 15 % of the original data withheld before augmentation, reached 0.85, with a mean absolute error (MAE) of 0.3598. These results highlight the effectiveness of data augmentation and targeted feature selection in enhancing the model's accuracy (Fig. 7f).

Here, we present an example of the results obtained for two FDA-

approved inhibitors of TgDHFR, pyrimethamine and trimethoprim, both used in *T. gondii* treatment (Table 2). The relative errors in the predicted pIC_{50} values compared to their experimentally measured pIC_{50} values were 3.35 % and 2.15 %, respectively, for pyrimethamine and trimethoprim. With an R^2 of 0.85, both predictions are within the model's expected accuracy, demonstrating reliable performance. Also, pyrimethamine exhibited higher ligand efficiency, binding efficiency index, and surface efficiency index than trimethoprim, confirming its superior inhibitory potency. These findings demonstrate that the theoretical predictions obtained in this paper, specifically the predicted pIC_{50} values, are in good agreement with the experimental data, validating the model's accuracy and reliability in assessing TgDHFR inhibitors.

4.4. Prediction of pIC_{50} for selected FDA drugs

In the next step, the predictive model obtained was used to scan the Food and Drug Administration (FDA) database to help find molecules that could be potentially efficient TgDHFR protein inhibitors. This database of 3451 drugs was sourced from PubChem[55]. Analysis of the FDA dataset showed that this database differs significantly from our training dataset, as evidenced by the analysis of key molecular properties, including atom composition, molecular weight, polar surface area, XLogP, heavy atom count, hydrogen bond donors, hydrogen bond acceptors, and the number of rotatable bonds (see Figs. S1, S2). These differences suggest that the FDA compounds occupy a different chemical space, which may affect the reliability of predictions made by our model. As such, careful consideration is required to assess the model's applicability to the FDA dataset.

First, we filtered the FDA dataset to retain only molecules with the same elemental composition as the training set, specifically those containing Br, I, Cl, H, O, P, S, F, N, and C. This ensures chemical consistency between the FDA compounds and the training data, minimizing discrepancies due to differing atomic compositions. Next, we excluded molecules with molecular weights exceeding 1000 Da to focus on compounds within a comparable size range. Finally, we applied an Isolation Forest algorithm with a 5 % contamination level to detect outliers in the training data. Using this model, we identified and filtered outliers in the FDA dataset, resulting in a refined set of 193 molecules more closely aligned with the systems in our training dataset (Fig. S3). Finally, pIC_{50} values were predicted for these molecules.

Based on our analysis of the predicted pIC_{50} values and key efficiency metrics, bisacodyl and etodolac are the most promising potential TgDHFR inhibitors with the highest predicted pIC_{50} of 7.43 and 7.20, respectively, and favorable ligand and binding efficiencies (Table 2, Fig. 8). Additionally, triamterene and lorlatinib show strong potential. Triamterene exhibits efficient binding and potent inhibition with a ligand efficiency (LE) of 0.49 and the highest binding efficiency index

(BEI) of 26.95. Its lipophilic ligand efficiency (LLE) 5.83 ensures a good balance between potency and lipophilicity, making it a strong candidate for drug development. Lorlatinib maintains potency with an LLE of 5.43 while having favorable lipophilicity, although its LE and BEI are slightly lower than Triamterene's. These efficiency metrics, especially LE, BEI, and LLE, are critical for evaluating the potential of these compounds as effective inhibitors.

5. Future work

To make informed decisions in drug development, it is essential to evaluate not only the inhibitory potential of drug candidates but also their safety profiles, including toxicity and potential side effects. A thorough assessment is necessary to ensure these compounds are suitable for therapeutic use, as adverse effects could limit their viability as TgDHFR inhibitors. From an experimental drug repurposing perspective, future research should prioritize molecular docking studies to screen and rank potential inhibitors of key *Toxoplasma gondii* targets, such as TgDHFR. Docking simulations will provide insights into binding affinities and interaction patterns with target proteins, which are critical for selecting promising candidates for further analysis [56]. Subsequently, molecular dynamics (MD) simulations should be employed to evaluate the stability and behavior of the inhibitors within the binding sites under dynamic physiological conditions, refining the selection process by focusing on compounds with favorable binding dynamics and stability [57].

Once promising inhibitors are identified through docking and MD simulations, an AI-driven approach should be used to predict their pharmacokinetic properties, including absorption, distribution, metabolism, excretion, and toxicity (ADMET) [58]. This computational step is crucial for filtering out compounds with undesirable pharmacokinetic profiles, ensuring that only inhibitors with favorable ADMET predictions progress to preclinical studies. This combined framework of docking, MD simulations, and AI-driven ADMET predictions can significantly improve the efficiency of drug repurposing efforts by focusing on both molecular interactions and drug-like properties, thereby reducing the time and resources required for experimental validation.

Additionally, in our next steps, we are planning to evaluate the performance of a bidirectional transformer-encoded fingerprint model, which has shown success in recent drug repurposing studies, such as opioid drug discovery [59]. This model, pre-trained with millions of SMILES sequences from ChEMBL, PubChem, and ZINC databases, could enhance the predictive power of our deep learning models. Incorporating this advanced method into our workflow will provide a robust comparison with our current DNN models, helping to validate our findings further and optimize future screening strategies [60].

Table 2

Two FDA-approved inhibitors used in *T. gondii* treatment and top ten FDA drugs with the highest predicted pIC_{50} against TgDHFR, and calculated LE, BEI, LLE, and SEI parameters.

Name	CID	pIC_{50} (predicted)	pIC_{50} (measured)	LE	BEI	LLE	SEI
Pyrimethamine*	4993	6.34	6.56 [35]	6.50	26.22	4.18	17.89
Trimethoprim*	5578	5.69	5.57 [35]	0.36	19.18	5.17	12.8
Bisacodyl	2391	7.43	—	0.38	20.55	3.63	11.34
Finerenone	60150535	7.35	—	0.36	19.42	4.95	6.68
Etodolac	3308	7.20	—	0.47	25.04	4.40	11.55
Ozenoxacin	9863827	7.09	—	0.36	19.50	2.99	8.59
Lorlatinib	71731823	6.93	—	0.32	17.06	5.43	6.30
Lemborexant	56944144	6.88	—	0.31	16.77	3.68	8.94
Triamterene	5546	6.83	—	0.49	26.95	5.83	5.25
Deferasirox	214348	6.76	—	0.33	18.11	2.96	6.20
Metolazone	4170	6.58	—	0.38	17.98	2.48	6.51
Methotrexate	126941	6.54	—	0.27	14.39	8.34	3.10

(*) FDA-approved inhibitors targeting TgDHFR (used in *Toxoplasma gondii* treatment). LE, BEI, LLE, and SEI were calculated using experimental pIC_{50} for these molecules.

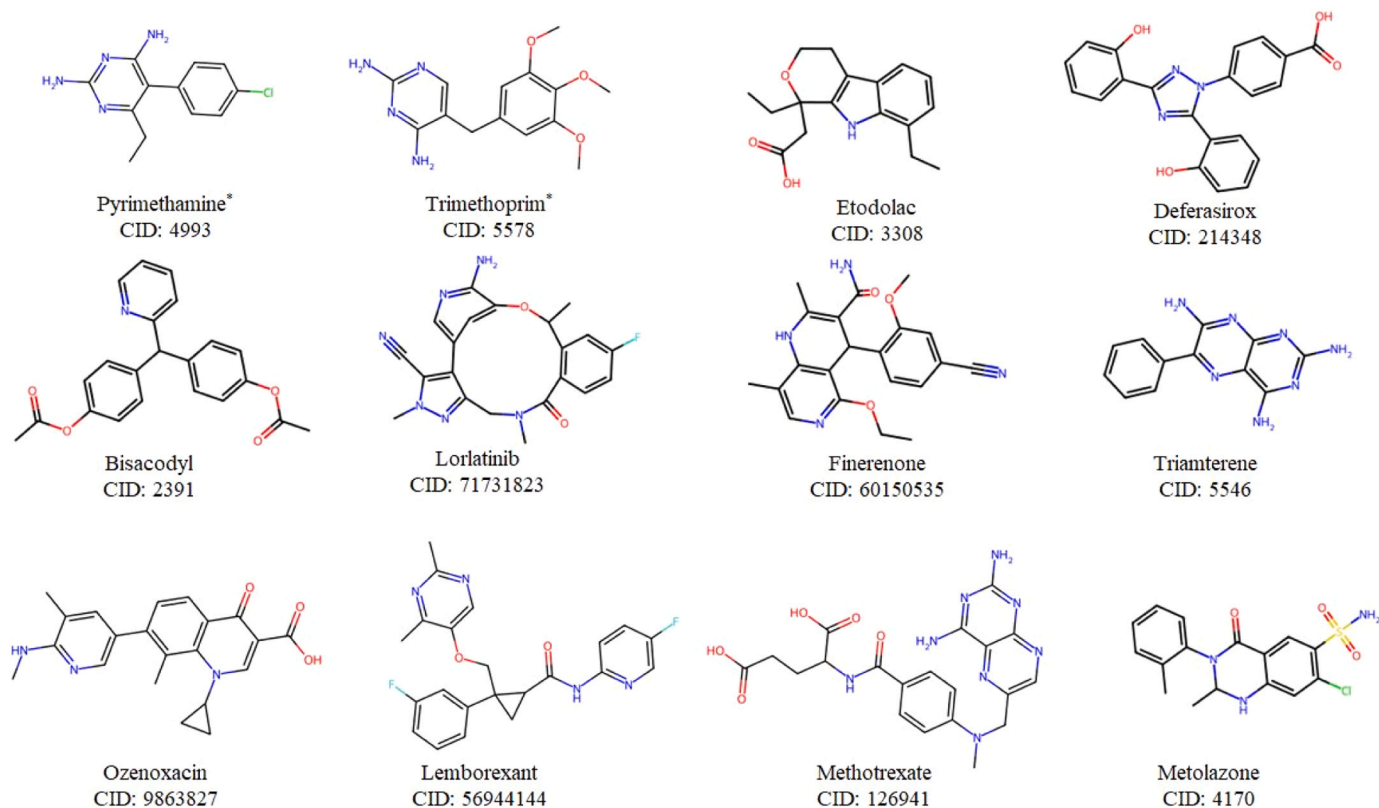


Fig. 8. 2D representations of the two FDA-approved inhibitors used in *T. gondii* treatment and the top 10 FDA drugs with the highest predicted pIC_{50} .

6. Conclusions

This study introduced a methodology to overcome the challenges of limited data in pIC_{50} prediction for TgDHFR inhibitors. We achieved significant performance improvements by combining targeted descriptor selection, Gaussian noise-based data augmentation, and an ensemble of deep neural networks. While elements of this approach have been used separately, this structured integration represents an innovative advancement in pIC_{50} prediction for drug discovery with limited datasets. This approach plays a crucial role in developing new treatments for *T. gondii*, addressing the need for improved therapies amid limited options and emerging drug resistance. The model exhibited strong predictive capabilities by increasing performance from an initial R^2 of 0.75–0.82 by selecting features with positive impact and further to 0.85 by employing data augmentation and ensemble DNN approaches.

Results identified 'AutocorrelationDescriptor' and 'ElectropologicalStateAtomTypeDescriptor' as key 2D descriptors, emphasizing the importance of spatial atomic properties and electronic environments. In the 3D category, 'RDFDescriptors' highlights the significance of atomic distance distributions in molecular interactions. For fingerprint descriptors, 'SMARTS Patterns' capture critical substructures, providing detailed insights into chemical composition. These findings demonstrate that combining structural, electronic, and substructural features is essential for precise molecular activity prediction, ensuring a comprehensive approach to evaluating potential inhibitors.

Additionally, the model's reliability was validated on two FDA-approved TgDHFR inhibitors, pyrimethamine and trimethoprim, commonly used to treat *T. gondii* infections. The relative errors between the predicted and experimental pIC_{50} values were 3.35 % and 2.15 %, respectively, demonstrating strong predictive performance and aligning with the model's expected accuracy. Furthermore, the model's ability to filter FDA-approved drugs based on their alignment with the training dataset further enhanced prediction reliability. By incorporating a pre-screening step to ensure data alignment, followed by sequential AI-

driven pIC_{50} prediction, our approach offers a more efficient and accurate drug screening process, mainly when dealing with limited datasets.

The results highlighted bisacodyl, etodolac, triamterene, and lorlatinib as promising candidates for TgDHFR inhibition, showing strong potential based on key molecular parameters. Furthermore, this model can serve as a filter model in generative AI pipelines, streamlining drug discovery by assessing the activity of generated compounds and selecting high-quality molecules for further development. This method can also be applied to other proteins, particularly those with limited datasets, making it a versatile tool for rare diseases and under-explored targets. This adaptability extends its utility beyond TgDHFR, enhancing its relevance for broader protein inhibition challenges.

Ethics approval and consent to participate

No human or animal subjects were used as part of this work.

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CRediT authorship contribution statement

Senanayake Ravithree D.: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Karimova Natalia V.:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Consent for publication

The authors give their consent for this manuscript to be published.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.aichem.2025.100084.

Data and Code Availability

The code developed for this work is uploaded to https://github.com/karimova/pIC50_paper.

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